

Why TRANSITION From NOR To NAND Flash Memory

While NOR has a fast read speed that is beneficial for quick software loading in embedded devices, it cannot meet the storage power of NAND. So, when should you make the shift? Read to know more

ONG CHUEN WEE
is Engineering
Manager at
Memory Marketing
Department, KIOXIA
Singapore

In the current market scenario, almost all the embedded systems are becoming smarter, resulting in the development of innovative smart TVs, smartwatches, and many other IoT devices every one to five years. Part of this increased smartness is due to fast communication protocols like 4G LTE, 5G, Wi-Fi, and GPON that have connected everything—no matter how far they are.

While all this brings added intelligence and richer user interface (UI) that makes our tasks a tad less difficult, consequently, it requires higher processing capacity, for which flash memory plays an essential role. Commonly used flash memory types in modern digital systems include NOR and NAND. Traditionally, NOR types are used in digital devices owing to their fast read capability with respect to a large firmware.

But NOR devices are quite expensive and consume a lot of power, which decreases in standby mode. Moreover, for applications requiring high write speed, NOR is not adequate. This is where NAND with its faster write speeds and higher memory capacity can make a difference.

So, before we dive deep into the technicalities of NAND, let us see the Table below for a quick comparison between NAND and NOR types.

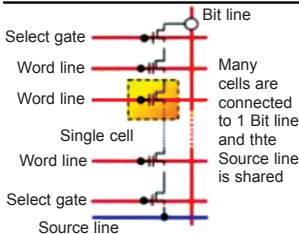
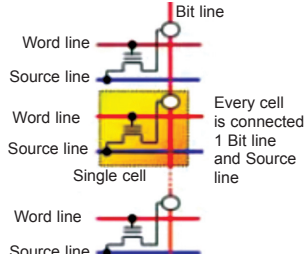
For the NAND flash, a single memory cell is at least 2.5 times smaller than the NOR flash, making NAND less expensive. To understand more about the internal structure of NAND and how successful transition can be made from NOR, let us look at the following.

NAND flash memory structure

In the diagram shown in Fig. 1 for 1-gigabit (1Gb) NAND flash memory, there are two data caches of 1 and 0. Below them, all are memory cells or arrays. For 1Gb serial NAND, we have a page size of 2048 bytes plus 64 bytes for spare. The structure is enabled with internal error code correction (ECC) that can be disabled as well.

When 64 pages are brought together, it becomes a single block, that is, 64 pages = 1 block. It comprises 128-kilobyte (kB) memory size plus 4kB for spare. Total 1024 blocks make up 1-gigabit (Gb), which equals 128 (or 125) megabytes (MB), as eight bits equal one byte.

NAND AND NOR COMPARISON

Specifications	NAND	NOR
Cell array	Single Source line connected to a single Bit line share many memory cells. Multiple Word lines together pass through the Bit line. 	Several Source lines connected to a single Bit line. Word lines above each Source line pass through the Bit line. 
Cell size	$2F \times 2F = 4F^2$	$5F \times 2F = 10F^2$
Average read speed	20.5MBps	61.5MBps
Average write speed	3.2MBps	0.47MBps